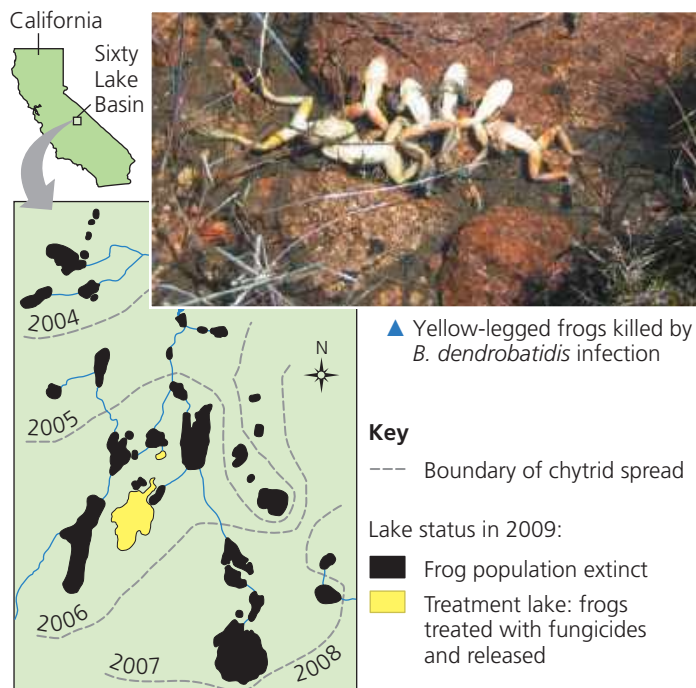


▼ **Figure 31.27 Amphibians under attack.** The number of yellow-legged frogs (*Rana muscosa*) plummeted after the chytrid *Batrachochytrium dendrobatidis* reached the Sixty Lake Basin area of California. In the years leading up to the chytrid's 2004 arrival, there had been more than 2,300 frogs in these lakes. By 2009, only 38 frogs remained; all the survivors were in two lakes (yellow) where frogs had been treated with a fungicide to reduce the chytrid's impact.



INTERPRET THE DATA Do the data depicted indicate that the chytrid caused or is correlated to the drop in frog numbers? Explain.

to massive die-offs (Figure 31.27). Field observations and studies of museum specimens show that *B. dendrobatidis* and *B. salamandrivorans* first appeared in amphibian populations shortly before their declines in Australia, Costa Rica, Germany, the United States, and other countries. Genetic analyses indicate that both *B. dendrobatidis* and *B. salamandrivorans* originated in Asia and spread from there via the commercial trade of frogs and salamanders.

The general term for an infection in an animal by a fungal parasite is **mycosis**. In humans, skin mycoses include the disease ringworm, so named because it appears as circular red areas on the skin. Most commonly, the ascomycetes that cause ringworm grow on the feet, causing the intense itching and blisters known as athlete's foot. Though highly contagious, athlete's foot and other ringworm infections can be treated with fungicidal lotions and powders.

Systemic mycoses, by contrast, spread through the body and usually cause very serious illnesses. They are typically caused by inhaled spores. For example, coccidioidomycosis is a systemic mycosis that produces tuberculosis-like symptoms

in the lungs. Each year, hundreds of cases in North America require treatment with antifungal drugs, without which the disease could be fatal.

Some mycoses are opportunistic, occurring only when a change in the body's microorganisms, chemical environment, or immune system allows fungi to grow unchecked. *Candida albicans*, for example, is one of the normal inhabitants of moist epithelia, such as the vaginal lining. Under certain circumstances, *C. albicans* can grow too rapidly and become pathogenic, leading to so-called "yeast infections." A related species, *C. auris*, has emerged as a global threat, often in healthcare facilities. Resistant to multiple antifungal drugs, this species can infect the bloodstream and cause life-threatening infections.

Practical Uses of Fungi

The dangers posed by fungi should not overshadow their immense benefits. We depend on their ecological services as decomposers and recyclers of organic matter. In addition, mushrooms are not the only fungi of interest for human consumption. Fungi are used to ripen Roquefort and other blue cheeses. Morels and truffles, the edible fruiting bodies of various ascomycetes, are highly prized for their complex flavors (see Figure 31.17). These fungi can sell for hundreds to thousands of dollars a pound. Truffles release strong odors that attract mammals and insects, which in nature feed on them and disperse their spores. In some cases, the odors mimic the pheromones (sex attractants) of certain mammals. For example, the odors of several European truffles mimic the pheromones released by male pigs, which explains why truffle hunters sometimes use female pigs to help find these delicacies.

Humans have used yeasts to produce alcoholic beverages and bread for thousands of years. Under anaerobic conditions, yeasts ferment sugars to alcohol and CO₂, which causes dough to rise. Only relatively recently have the yeasts involved been separated into pure cultures for more controlled use. The yeast *Saccharomyces cerevisiae* is the most important of all cultured fungi (see Figure 31.7). It is available as many strains of baker's yeast and brewer's yeast.

Many fungi have great medical value as well. For example, a compound extracted from ergots is used to reduce high blood pressure and to stop maternal bleeding after childbirth. Some fungi produce antibiotics that are effective in treating bacterial infections. In fact, the first antibiotic discovered was penicillin, made by the ascomycete mold *Penicillium*. Other examples of pharmaceuticals derived from fungi include cholesterol-lowering drugs and cyclosporine, a drug used to suppress the immune system after organ transplants.

Fungi also figure prominently in basic research. For example, the yeast *Saccharomyces cerevisiae* is used to study the